

Project Performance Audit & Cost Report

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Executive Summary

This is a Commercial building with 4 structural divisions, including 2 bedrooms, 2 bathrooms, 1 kitchen, and 1 hall. Based on the uploaded blueprint, the estimated construction cost is ₹3,08,26,313 using Standard quality specifications. The estimated completion time is 7 months.

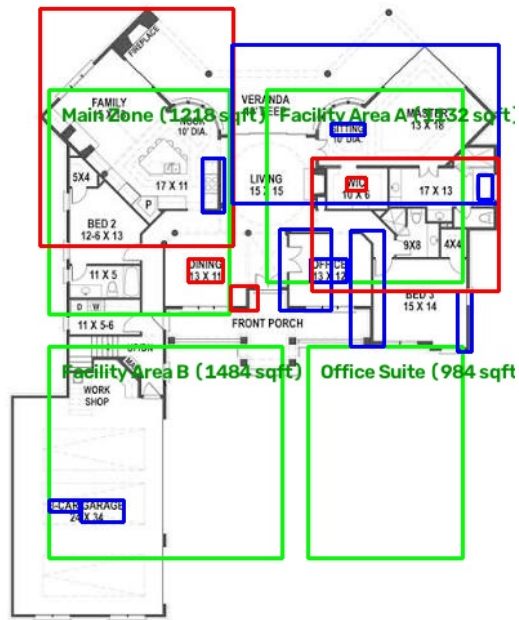
Project Details & Assigned Personnel

Project Name:	home	Client Name:	aswin
Location:	coimbatore	Building Type:	Commercial
Floors:	2	Material Quality:	Standard
Assigned PM:	Not Assigned	Assigned Engineer:	Not Assigned

AI Blueprint Layout Analysis

Structural Division	Detected Quantity	Classification
Bedrooms / Cabin Areas	2 Zones	Enclosure Space
Bathrooms / Plumbing	2 Zones	Sanitary Block
Kitchen / Cafeteria	1 Zone	Services Block
Hall / Reception Zone	1 Zone	Common Area
Dining / Pantry Area	0 Zone	Amenity space
Balconies	2 Zones	Open Air Extension
Parking Slots	1 Zone	Utility
Staircase / Lift Core	1 Zone	Vertical Circulation
Doors Identified	8 Units	Apertures / Access
Windows Identified	12 Units	Apertures / Ventilation
Total Built-up Area	9,636 sqft	Calculated Area

Project Blueprint Layout Drawing Reference:



AI Material Quantity & Sourcing Schedule

Material Item	Estimated Qty	Unit	Total Cost (INR)
Cement	3,855	bags	₹14,64,900
Steel Rebar	38.54	tons	₹23,89,480
Bricks	211,992	pcs	₹16,95,936
Sand	771	tons	₹9,25,200
Blue Metal Aggregate	579	tons	₹8,10,600
Gravel / Soling	386	tons	₹3,86,000
Ready-Mix Concrete	290	cum	₹13,05,000
Paint	1,542	liters	₹4,00,920
Flooring Tiles	11,082	sqft	₹7,20,330
Electrical Conduit & Wiring	1	lump-sum	₹8,67,240
Plumbing Pipes & Fixtures	1	lump-sum	₹7,70,880
Roofing/Slab Concrete reinforcement	1	lump-sum	₹12,52,680
Doors	8	units	₹48,000
Windows	12	units	₹48,000
Labor Sourcing Requirement	8,191	man-days	₹45,05,050

Financial Summary & Cost Breakdown

Cost Category	Estimated Cost (INR)	Cost Category	Estimated Cost (INR)
Materials Base Cost	₹1,30,85,166	Labor Cost	₹45,05,050
Equipment & Machinery	₹19,27,200	Electrical Works	₹8,67,240
Plumbing & Sewerage	₹7,70,880	Painting & Coatings	₹4,00,920
Roofing Works	₹12,52,680	Finishing Works	₹12,17,250
Government Approvals	₹1,50,000	Miscellaneous Expenses	₹7,03,609
Contingency Amount (5%)	₹12,44,000	GST (18% applicable)	₹47,02,319
Grand Total Estimate	₹3,08,26,313		

Construction Timeline & Project Schedule

Construction Phase / Stage	Estimated Duration (Weeks)
Site Preparation	2
Foundation	4
Column Work	3
Wall Construction	4
Roofing	3
Electrical Work	2
Plumbing	2
Flooring	3
Painting	2
Finishing	3
Overall Estimated Duration	7 Months

AI Construction Risk Scorecard

Risk category	Level	Telemetry Diagnosis	Mitigation Recommendation
Delay Risk	Low	Slight logistical supply delay risks or approval buffer slip.	Maintain a 14-day schedule buffer.
Budget Risk	Low	Market steel price volatility.	Procure major materials at project kickoff.
Safety Risk	Medium	Scaffolding falls and heavy vehicle movements.	Enforce mandatory hard-hats and harness usage.
Material Sourcing	Medium	Sourcing delays for premium finishes.	Source from multiple supplier agreements.
Weather Risk	Low	Seasonal monsoon rainfall interruptions.	Perform roofing concrete pours before peak rainy season.

AI Sourcing & Value Optimization Recommendations

- Suggested Workforce: 78 active workers (skilled & unskilled)
- Suggested Machinery: Concrete Mixer Machine, Slab Vibrator, JCB Excavator
- Cost Optimization: Optimize cut-lists for steel rebar to minimize waste below 5%.
- Construction Sequence: Run electrical conduiting concurrently with brick laying partitions.
- Material Sourcing Improvements: Use fly-ash blended OPC cement for columns foundation slab to lower footprint.
- Safety Program: Conduct safety toolboxes each morning for working-at-height laborers.

Final Conclusion: This cost and quantity report is generated based on standard mechanical layouts and area bounding boxes detected from the building blueprint. The quantities and rates comply with current Central Public Works Department (CPWD) schedule indices in India and are ready for commercial execution review.

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Project Manager

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